RÉCAPITULATIF DE LA CONFIGURATION FINALE

Document créé dans le cadre de l'épreuve E6 du BTS SIO SISR

Élément	Configuration
FortiGate port4	192.168.44.1/30
Switch source SPAN	Fa0/1 (both) - trunk 10,20,30,99
Switch destination SPAN	Fa0/44 (ingress)
PC portable	192.168.44.2 255.255.255.252
Wireshark	Colonnes et filtres paramétrés.

FG-300D-BTS :

Installation du Fortigate 300D, voir la documentation de via Tftpd64.

Après l'installation. Supplément pour le projet Wireshark. *(sert à tester la connectivité du Fortigate depuis le pc portable Wireshark afin d'assurer la couche 2 du Fortigate).*

Bash

```bash
config system interface
    edit "port4"
        set mode static
        set ip 192.168.44.1 255.255.255.252
        set allowaccess ping https ssh
        set role lan
        set device-identification disable
        set alias "Wireshark"
        set description "Interface Management VLAN 444 /30"
    next
end
```

Configuration FG-300D-BTS des règles par-feu Baie D.

Name	From	To	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
VLAN99_to_VLAN10	VLAN99	VLAN10	all	all	always	ALL	ACCEPT	Disabled	SSL no-inspection	UTM	960 B
VLAN99_to_VLAN20	VLAN99	VLAN20	all	all	always	ALL	ACCEPT	Disabled	SSL no-inspection	UTM	8.48 MB
VLAN99_to_VLAN30	VLAN99	VLAN30	all	all	always	ALL	ACCEPT	Disabled	SSL no-inspection	UTM	72.65 MB
VLAN10_to_DHCP	VLAN10	VLAN20	all	WinServ	always	DHCP DNS	ACCEPT	Disabled	SSL no-inspection	UTM	65.00 kB
DHCP_to_VLAN10	VLAN20	VLAN10	WinServ	all	always	DHCP	ACCEPT	Disabled	SSL no-inspection	UTM	0 B
VLAN30_to_DHCP	VLAN30	VLAN20	all	WinServ	always	DHCP DNS	ACCEPT	Disabled	SSL no-inspection	UTM	6.22 MB
DHCP_to_VLAN30	VLAN20	VLAN30	WinServ	all	always	DHCP	ACCEPT	Disabled	SSL no-inspection	UTM	0 B
VLAN10_to_VLAN20_HTTPS	VLAN10	VLAN20	all	all	always	HTTPS HTTP	ACCEPT	Disabled	SSL no-inspection	UTM	37.54 kB
VLAN10_to_VLAN30_HTTPS	VLAN10	VLAN30	all	all	always	HTTPS HTTP	ACCEPT	Disabled	SSL no-inspection	UTM	67.52 kB
OpenVPN_WAN_to_DMZ	WAN (port1)	VLAN30	all	OpenVPN_VIP	always	OpenVPN	ACCEPT	Disabled	SSL no-inspection	UTM	3.00 MB
VPN_to_VLAN10	VLAN30	VLAN10	réseau_VPN	VLAN10_Subnet	always	ALL	ACCEPT	Enabled	SSL no-inspection	UTM	0 B
VLAN10_to_WAN	VLAN10	WAN (port1)	all	all	always	DNS HTTP HTTPS ALL_ICMP	ACCEPT	Enabled	SSL no-inspection	UTM	7.97 MB
VLAN20_to_WAN	VLAN20	WAN (port1)	all	all	always	DNS HTTP HTTPS ALL_ICMP	ACCEPT	Enabled	SSL no-inspection	UTM	438.59 MB
VLAN30_to_WAN	VLAN30	WAN (port1)	all	all	always	DNS HTTP HTTPS NTP ALL_ICMP	ACCEPT	Enabled	SSL no-inspection	UTM	13.17 GB
VLAN99_to_WAN	VLAN99	WAN (port1)	all	all	always	HTTP HTTPS ALL_ICMP	ACCEPT	Enabled	SSL no-inspection	UTM	6.18 GB

Vue synthétique .xlsx des règles par-feu Baie D.

	→ Destination				
Source →	**VLAN10**	**VLAN20**	**VLAN30**	**VLAN99**	WAN
VLAN10 – Client		**DHCP**, HTTP,DNS	HTTP, DNS	×	HTTP/HTTPS/DNS/ICMP
VLAN20 – Serveurs				×	HTTP/HTTPS/DNS/ICMP
VLAN30 – DMZ	(VPN)	**DHCP**, HTTP, DNS		×	HTTP/HTTPS/DNS/ICMP/NTP
VLAN99 – Management	ALL	ALL	ALL		HTTP/HTTPS/ ICMP
WAN (Établissement)	×	×	OpenVPN (VIP)	×	

Ajouts supplémentaires non réalisés dans le cadre du projet.

```
Config system interface
edit “Port4”
set device-identification enable.
```

Name	VLAN444_to_WAN
Incoming Interface	Wireshark (port4)
Outgoing Interface	WAN (port1)
Source	VLAN44_subnet
Destination	all
Schedule	always
Service	ALL
Action	ACCEPT / DENY
Inspection Mode	Flow-based / Proxy-based

Règles par-feu pour sortir sur le WAN et/ou autorisé VLAN20_to_VLAN444. Pour seulement avoir le service DNS du windows serveur.

SW-L3-BTS :

Supplément pour le projet Wireshark. (sert à copier le trafic du port fa0/1 vers la destination fa0/44).

```bash
enable
configure terminal
!
! 1. Réinitialisation par sécurité
no monitor session 1
!
! 2. Préparation du port de destination (Écoute Wireshark)
interface fa0/44
 description *** PC ANALYSE WIRESHARK ***
 no shutdown
 exit
!
! 3. Configuration de la session SPAN (Port Mirroring)
monitor session 1 source interface fa0/1 both
monitor session 1 destination interface fa0/44 encapsulation dot1q
!
end
write memory
```

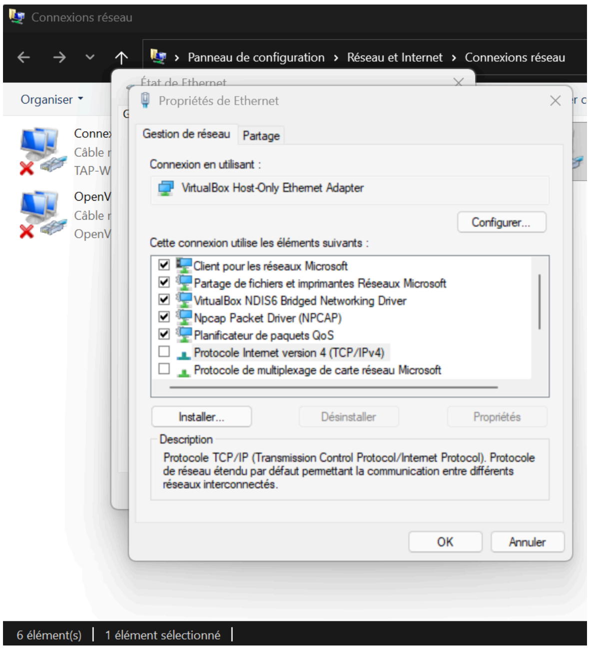
Configuration du SW-L3-BTS Baie D.

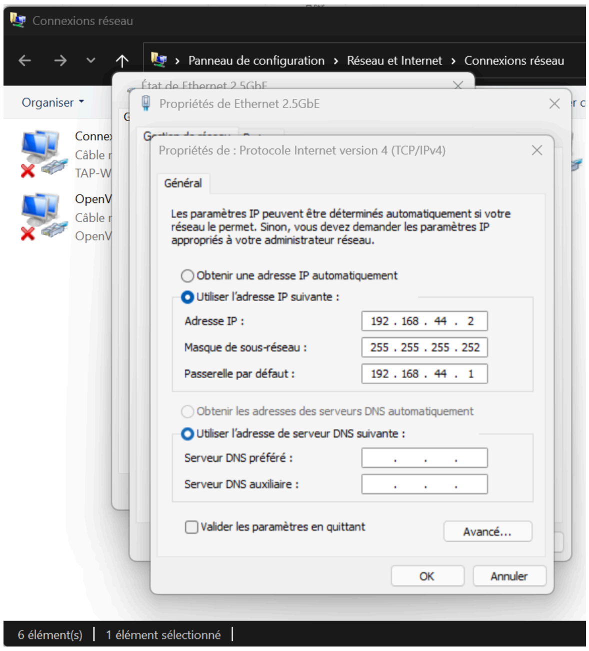
```
SW-L3-BTS#sh run int fa0/1
Building configuration...

Current configuration : 196 bytes
!
interface FastEthernet0/1
 description TRUNK_FORTIGATE
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 spanning-tree portfast trunk
end

SW-L3-BTS#sh run int fa0/17
Building configuration...

Current configuration : 228 bytes
!
interface FastEthernet0/17
 description TRUNK_PROXMOX
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 spanning-tree portfast trunk
end

SW-L3-BTS#sh run int fa0/44
Building configuration...

Current configuration : 112 bytes
!
interface FastEthernet0/44
 description Pc Wireshark
 switchport access vlan 444
 switchport mode access
end
```

```
SW-L3-BTS#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/37, Fa0/38, Fa0/39, Fa0/40
                                                Fa0/41, Fa0/42, Fa0/43, Fa0/45
                                                Fa0/46, Fa0/47, Fa0/48, Gi0/1
                                                Gi0/2
10   client                           active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16
20   srvinterne                       active    Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
30   dmz                              active    Fa0/25, Fa0/26, Fa0/27, Fa0/28
                                                Fa0/29, Fa0/30, Fa0/31, Fa0/32
99   management                       active    Fa0/33, Fa0/34, Fa0/35, Fa0/36
444  Analyse_Wireshark                active    Fa0/44
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2

SW-L3-BTS#sh mon
SW-L3-BTS#sh monitor
Session 1
---------
Type              : Local Session
Source Ports      :
    Both          : Fa0/1
Destination Ports : Fa0/44
    Encapsulation : DOT1Q
          Ingress: Disabled


SW-L3-BTS#
```

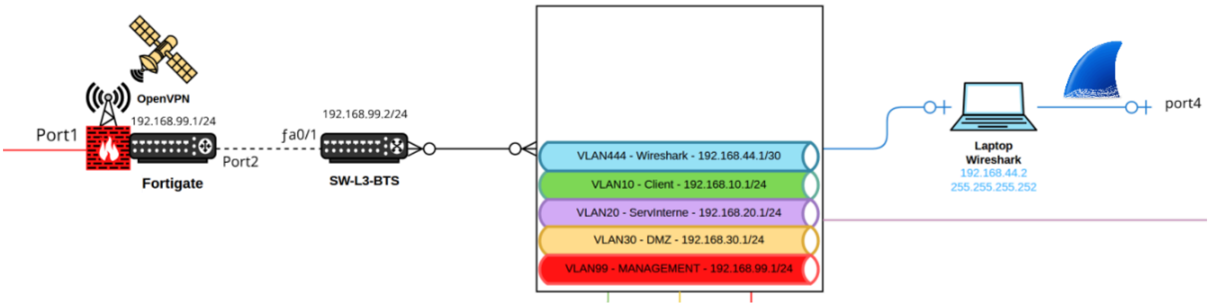
Pc portable :

Configuration des cartes réseaux. (fa0/44 en simplex vers le port Adaptateur Ethernet. Puis port4 du Fortigate vers le pc portable en 192.168.44.0/30).





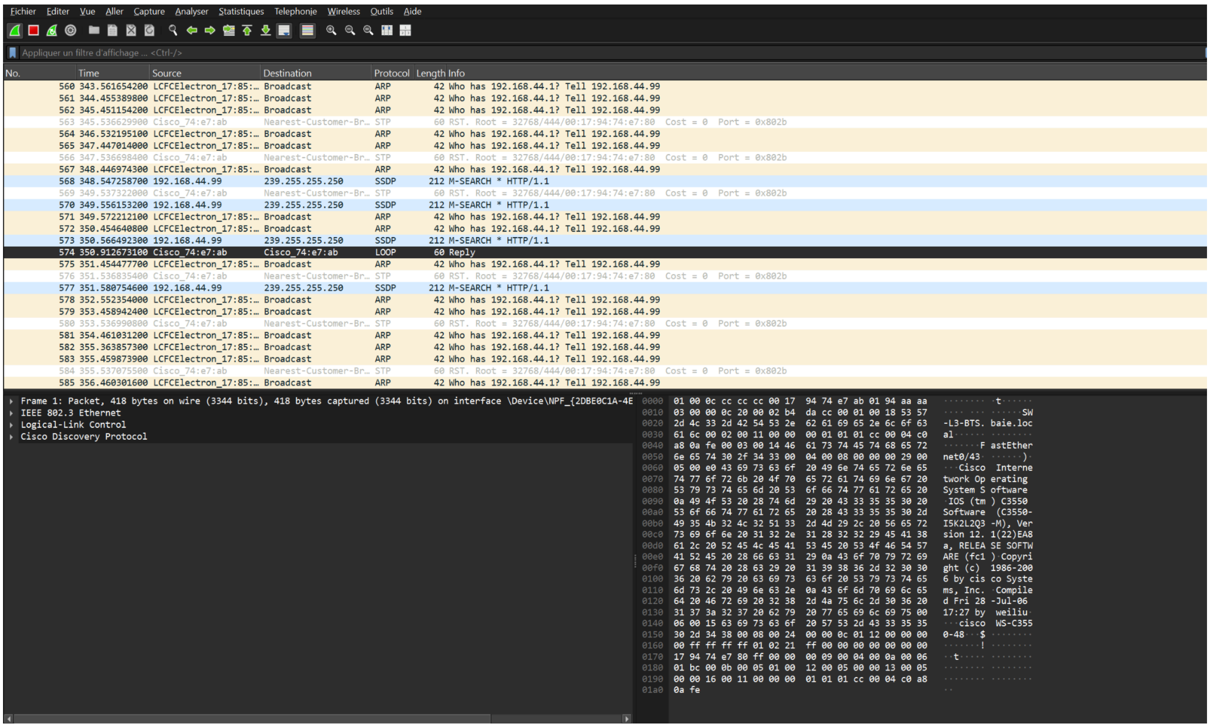
Vue de l'infrastructure réseau Wireshark ci-dessous :

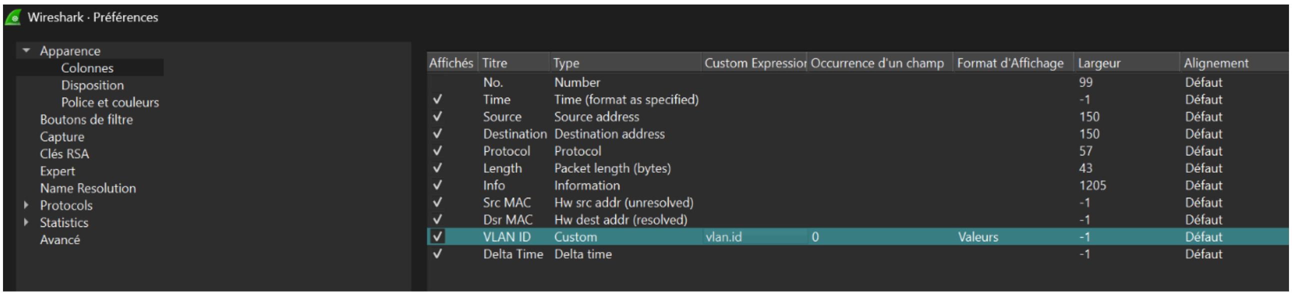


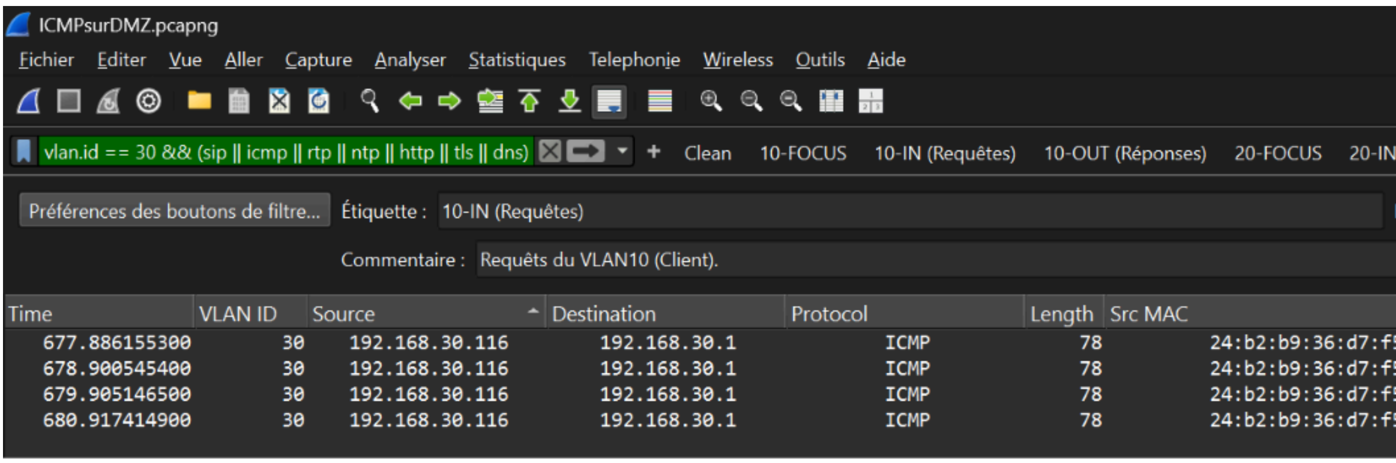
Le lien Ethernet 2.5GbE est configurable à tout moment pour la bascule vers VLAN99 (Management). En cas de problème détecté sur le trafic réseau.

Wireshark :

Configuration de l’interface Wireshark pour une bonne visibilité.







Le pc portable Analyse Wireshark **VLAN444** permet ainsi de vérifier que depuis le vlan 30 DMZ sur un poste DHCP dans cet exemple. Les protocoles reçus et envoyés arrivent bien jusqu’à la passerelle par défaut.